Java Notes: Shallow Copy vs Deep Copy

Shallow Copy

A shallow copy creates a new object, but the referenced objects inside it are shared. If one object modifies a shared referenced object, the change is visible through the other object as well.

Key Point: Reference variables point to the same nested objects.

Deep Copy

A deep copy creates a new object and also creates copies of all referenced objects. Changes made through one copy do not affect the other copy.

Key Point: All nested objects are copied independently.

Comparison

Shallow Copy: Faster, uses less memory, but shares referenced objects.
Deep Copy: Uses more memory and time, but provides complete independence.

Interview Definition

Shallow Copy copies object fields as-is. Reference fields still point to the same objects.
Deep Copy creates independent copies of the object and all nested objects.